Solar-Powered LoRa Sensor

Temco Controls 2026-05-27

SSL-1

Overview

The Solar-Powered LoRa Sensor is a field-deployable environmental node from Temco Controls. Energy is harvested from a compact solar panel, stored in a supercapacitor bank, and regulated to a 3.3 V system rail for always-on sensing and duty-cycled LoRa uplinks. High-draw peripherals (mmWave radar, CO_2) are gated so the harvester can cold-start and recover under low light.

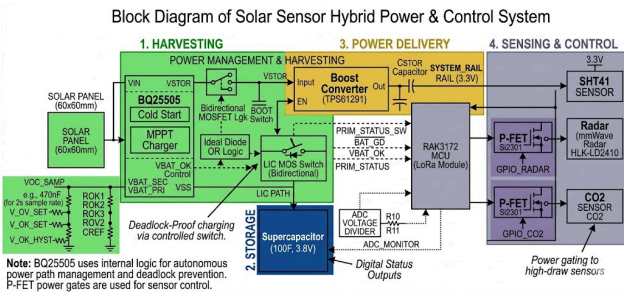
Source hardware and firmware live in the Solar-Powered-Lora-Sensor project. Copied reference assets for this datasheet build:

- `assets/block-diagram.jpg` — system block diagram
- `assets/block-diagram.html` — interactive diagram export
- `assets/schematic-rev2.pdf` — Rev2 schematic
- `assets/bom.csv` — bill of materials

Electrical specifications

Parameter	Value
System rail	3.3 V DC (SYSTEM_RAIL)
Energy harvester	BQ25505 — MPPT, cold start, path management
Boost regulator	TPS61291 ⟶ 3.3 V
Storage	100 F supercap (LIC path, 3.8 V class)
Solar input	60 × 60 mm panel ⟶ BQ25505
LoRa MCU	RAK3172 (LoRaWAN)
Always-on sensor	SHT41 — temperature / RH
Gated loads	mmWave radar, CO_2 (P-FET on 3.3 V)
Architecture	Hybrid — harvest + supercap OR-ing

Block diagram



Hybrid power and sensor topology — see `assets/block-diagram.html` and `assets/schematic-rev2.pdf`.

Power architecture

Stage	Function
Solar 60×60 mm	Input to BQ25505 harvester
BQ25505	Cold start, MPPT, VBAT_OK / path management

Stage	Function
100 F supercap	LIC storage; ideal-diode OR with primary path
TPS61291	Boost to regulated 3.3 V SYSTEM_RAIL
RAK3172	LoRaWAN MCU — always powered from system rail
SHT41	Environmental sensing — always on at 3.3 V
Gated radar / CO_2	High inrush loads enabled only when storage is healthy

Reference assets

Bundled under `examples/solar-lora-sensor/assets/` for offline builds:

Parameter	Value
Block diagram (image)	`block-diagram.jpg`
Block diagram (HTML)	`block-diagram.html`
Schematic Rev2	`schematic-rev2.pdf`
BOM	`bom.csv`

Notes

- Text and layout: edit this `.typ` file.
- Manifest: `datasheet.yaml` maps `source_project` and asset paths for automation.
- Re-sync assets from `Solar-Powered-Lora-Sensor` when the PCB or BOM changes.